Introduction to Machine Learning

What is Machine Learning?

Machine Learning is a branch of Artificial Intelligence that allows systems to learn and improve from experience without being explicitly programmed.

Types of Machine Learning

Supervised Learning uses labeled data to train models and make predictions on new data.

Unsupervised Learning finds hidden patterns and structures in unlabeled data.

Reinforcement Learning trains agents to make decisions by rewarding correct actions.

Common Algorithms

Popular ML algorithms include Linear Regression, Decision Trees, Random Forests, Support Vector Machines, and K-Nearest Neighbors.

Neural Networks are inspired by the structure of the human brain and are used in deep learning tasks.

Applications

Machine Learning is widely used in image recognition, natural language processing, recommendation systems, fraud detection, and autonomous vehicles.